

DAY 18 METHOD COMPARISON

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Group:- C2

MODEL / METHOD COMPARISON

METHODS COMPARED

- HISTOGRAM EQUALIZATION
- CLAHE
- GAMMA CORRECTION
- WHITE BALANCE
- REFINED CLAHE-3
- CNN

OBSERVATION:

CNN ACHIEVED THE HIGHEST PSNR AND SSIM, WHILE REFINED CLAHE-3 ACHIEVED THE HIGHEST EDGE F1.

<u>Method</u>	<u>PSNR</u>	<u>SSIM</u>	<u>Edge F1</u>
Histogram Equalization	8.0800	0.2225	0.0573
CLAHE	10.9900	0.4346	0.0572
Gamma Correction	10.9300	0.4821	0.0416
White Balance	10.4400	0.4611	0.0416
Refined CLAHE-3	11.4000	0.3995	0.0688
CNN	14.4692	0.5619	0.0259

PERFORMANCE ANALYSIS, LIMITATIONS AND FAILURE PATTERNS

PERFORMANCE

- **CNN ACHIEVED THE HIGHEST PSNR = 14.4692.**
- **CNN ACHIEVED THE HIGHEST SSIM = 0.5619.**
- **REFINED CLAHE-3 ACHIEVED THE HIGHEST EDGE F1 = 0.0688.**
- **GAMMA CORRECTION ACHIEVED THE HIGHEST SSIM AMONG THE BASIC CLASSICAL METHODS.**
- **HISTOGRAM EQUALIZATION PRODUCED THE LOWEST PSNR AND SSIM AMONG THE COMPARED METHODS.**

DATA LIMITATIONS

- **DATASET V1 CONTAINS 312 VALID PAIRED IMAGES.**
- **THE RELATIVELY SMALL DATASET CAN LIMIT GENERALIZATION.**
- **UNDERWATER SCENES CAN HAVE DIFFERENT LEVELS OF BRIGHTNESS, CONTRAST AND COLOUR DEGRADATION.**

FAILURE PATTERNS

- **CLASSICAL METHODS MAY IMPROVE VISIBILITY BUT CAN INTRODUCE OVER-ENHANCEMENT.**
- **STRONG ENHANCEMENT MAY PRODUCE ARTIFACTS OR GRAIN.**
- **CNN ACHIEVED BETTER PSNR/SSIM BUT ITS EDGE F1 WAS LOWER THAN THE CLASSICAL CANDIDATES.**
- **THIS SHOWS THAT DIFFERENT METRICS MEASURE DIFFERENT ASPECTS OF ENHANCEMENT QUALITY.**

SHORTLISTED CANDIDATES

CANDIDATE 1 — CNN

RESULTS:

- PSNR = 14.4692
- SSIM = 0.5619
- EDGE F1 = 0.0259

REASON:-

CNN ACHIEVED THE BEST PSNR AND SSIM AMONG ALL TESTED METHODS, MAKING IT THE STRONGEST CANDIDATE FOR FURTHER INVESTIGATION.

CANDIDATE 2 — REFINED CLAHE-3

PARAMETERS:

- CLIP LIMIT = 3.0
- TILE GRID = 8×8
- PSNR = 11.4000
- SSIM = 0.3995
- EDGE F1 = 0.0688

REASON:-

REFINED CLAHE-3 ACHIEVED THE BEST EDGE F1 AND PROVIDES A STRONG CLASSICAL REFERENCE FOR COMPARISON.



THANK YOU